FACEBOOK-CAMBRIDGE ANALYTICA DATA PRIVACY SCANDAL

CASE STUDY REPORT

submitted by

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1 Introduction

In the modern digital age, social media platforms have become an important part of everyday life. These platforms allow people to communicate, share information, and interact with others across the world. At the same time, they collect large amounts of personal data such as user profiles, interests, online activities, and social connections. While this data helps companies improve services and provide personalized content, it also raises serious concerns about data privacy and security. One of the most widely used social networking platforms is Facebook, which connects billions of users globally. Facebook collects various types of information through posts, likes, comments, and interactions on the platform. This information is stored in large databases and may also be shared with third-party developers through application programming interfaces (APIs) to create external applications and services. However, if proper security measures and privacy controls are not implemented, such access to user data can lead to misuse. In many cases, users are not fully aware of how their information is collected, stored, and shared with other organizations. This creates risks related to data privacy, unauthorized data access, and unethical use of personal information. A major example of such misuse is the Facebook–Cambridge Analytica data privacy scandal of 2018, where personal data from millions of Facebook users was collected through a third-party application and shared with Cambridge Analytica. This incident raised global concerns about user consent, data protection, and the importance of strong database security practices.

2 Background of the Organization

Facebook is one of the largest social networking platforms in the world. It was founded in 2004 by Mark Zuckerberg along with his college roommates at Harvard University. The platform was initially created to help students connect and commu1 nicate online, but it quickly expanded to users across the globe. Over time, Facebook became a major technology company providing services such as social networking, messaging, photo sharing, and online advertising. Facebook allows users to create personal profiles, connect with friends, share posts, photos, and videos, and interact through comments and messages. The platform collects various types of user information including personal details, interests, and online behavior. This data helps Facebook deliver personalized content and targeted advertisements to users. The company stores this information in large databases and uses advanced technologies to analyze user behavior and improve its services. Another organization involved in the scandal is Cambridge Analytica, a British political consulting and data analytics company. Cambridge Analytica specialized in data analysis and strategic

communication for political campaigns. The company used data-driven techniques to understand voter behavior and influence public opinion through targeted political advertising. The involvement of these organizations in the data privacy scandal raised serious concerns about how companies collect, store, and use personal information, highlighting the importance of strong data protection policies and ethical use of user data.

3 Problem or Violation Description

3.1 The Vulnerability

The Facebook–Cambridge Analytica scandal occurred mainly due to weaknesses in the data access policies of Facebook. At that time, Facebook allowed third-party applications to access large amounts of user data through its platform APIs. When users installed an application and granted permission, the app could also collect information about the user's friends. This policy created a vulnerability where personal data could be gathered from millions of users without their direct consent. The lack of strict monitoring and control over third-party developers increased the risk of misuse of user information.

3.2 Attack / Misuse

The misuse of data was carried out through a personality quiz application called "This Is Your Digital Life," developed by Aleksandr Kogan. Users who installed the app unknowingly allowed it to collect their profile data. Due to Facebook's datasharing policies at that time, the application also collected data from the friends of those users. The collected data was later transferred to Cambridge Analytica, which used it to build psychographic profiles and design targeted political advertisements.

3.3 Data Exposed

The data collected included personal profile details, Facebook likes, interests, social connections, and behavioral information. Although only around 270,000 users installed the application, data from nearly 87 million Facebook users was obtained through their friend networks. This data was used to analyze user personalities and influence voter behavior during political campaigns.

3.4 Detection and Disclosure

The misuse of data became public in 2018 after investigations by journalists and whistleblowers revealed that Cambridge Analytica had improperly obtained and

used Facebook user data. Reports by media organizations exposed the scale of the data misuse, leading to global criticism and government investigations. The incident forced Facebook to review its data privacy policies and highlighted the importance of stronger data protection and regulation.

4 Analysis: Laws and Principles Violated

4.1 Data Protection Act (1998)

The Data Protection Act 1998 is a law in the United Kingdom that regulates how personal data should be collected, stored, and used by organizations. The law requires that personal data must be processed fairly, lawfully, and only for specific purposes. In this case, the data collected through a personality quiz application was originally presented as being for academic research. However, the data was later transferred to Cambridge Analytica and used for political advertising and voter profiling. This misuse of data violated the principles of lawful and fair data processing.

4.2 Federal Trade Commission Act

The Federal Trade Commission (FTC) Act protects consumers from unfair or deceptive business practices. Under this law, companies must clearly inform users about how their data will be collected and used. Facebook had assured users that their personal data would be protected and not shared without permission. However, the data collected through third-party applications was later shared and used in ways that users did not expect. This was considered a deceptive practice and led to investigations and penalties against Facebook by the Federal Trade Commission.

4.3 General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR) is a European Union law designed to protect the personal data and privacy of individuals. It requires organizations to collect personal data only with the clear and informed consent of users and to explain how the data will be used. In the Facebook–Cambridge Analytica scandal, personal information from millions of users was collected without their proper consent. Many users were unaware that their data and the data of their friends would be accessed through the third-party application. This violated GDPR principles such as transparency, lawful processing, and protection of personal data.

4.4 Data Ethics Principles Violated

Data ethics requires organizations to collect and use personal information responsibly, transparently, and with user consent. In this case, personal data from millions of users of Facebook was collected through a quiz application created by Aleksandr Kogan. The data was later shared with Cambridge Analytica and used for political profiling and targeted advertisements. Users were not properly informed about this use of their data, which violated ethical principles of transparency, responsibility, and respect for user privacy.

5 Impact and Consequences

The Facebook–Cambridge Analytica scandal had major impacts on individuals, organizations, and governments worldwide. The misuse of personal data from millions of users on Facebook created serious concerns about data privacy, security, and ethical use of personal information. The incident exposed weaknesses in how social media platforms manage and protect user data. One of the major consequences was the loss of public trust in Facebook. Many users became worried about how their personal information was collected and shared with third-party organizations. As a result, people began paying more attention to privacy settings and data protection. The scandal also led to legal investigations and financial penalties. Governments and regulatory authorities investigated Facebook's data practices. The company faced large fines and was required to improve its privacy policies and security measures. Another important consequence was the increase in global awareness about data privacy. The incident encouraged governments to implement stronger data protection laws and regulations to protect user information. 4 Additionally, the scandal highlighted the risks of third-party application access to user data and forced companies to introduce stricter controls over developer access. Finally, the case raised concerns about political manipulation through data analytics, showing how personal data could be used to influence voter behavior and public opinion during elections.

6 Actions Taken and Penalties

6.1 Congressional Hearing Investigation

After the scandal became public, Mark Zuckerberg was called to testify before the United States Congress in 2018. During the hearing, lawmakers questioned Facebook about how user data was collected, shared with third-party applications, and protected. The investigation aimed to determine whether Facebook had violated privacy rules and to understand how such large-scale misuse of personal data occurred.

6.2 Federal Trade Commission Investigation

The Federal Trade Commission conducted a detailed investigation into Facebook's data privacy practices. The investigation examined whether Facebook had misled users about how their personal information was protected and shared with thirdparty developers. This investigation later led to significant penalties and stronger privacy requirements for the company

6.3 Platform Policy Changes by Facebook

Following the scandal, Facebook introduced several changes to improve user privacy. The company restricted third-party access to user data, strengthened data protection policies, and provided users with better control over their privacy settings to prevent similar incidents in the future.

6.4 Federal Trade Commission (FTC) Fine

The Federal Trade Commission investigated the data privacy practices of Facebook after the Cambridge Analytica scandal. The investigation found that Facebook failed to properly protect the personal data of its users and allowed third-party applications to misuse this information. As a result, in 2019 the FTC imposed a 5 billion fine on Facebook.This became one of the largest penalties ever issued for violating user privacy.

6.5 Information Commissioner's Office (ICO) Fine

The Information Commissioner's Office in the United Kingdom also investigated the case and found that Facebook had not adequately safeguarded the personal data of its users. Under the Data Protection Act, the ICO fined Facebook £500,000, which was the maximum fine allowed under the law at that time.

7 Recommendations and Preventive Measures

7.1 Stronger Data Access Control

Organizations such as Facebook should implement strict access control mechanisms to regulate how third-party developers access user data. Only authorized applications should be allowed to access specific types of information required for their functions. Limiting access to essential data reduces the chances of misuse and helps protect sensitive user information from being collected unnecessarily

7.2 Clear User Consent Policies

Companies must ensure that users are clearly informed about how their data will be collected, stored, and used. Applications should request explicit permission before accessing personal data. Users should also be able to review and withdraw permissions at any time. Transparent consent policies help build trust between users and organizations.

7.3 Regular Security Audits and Monitoring

Organizations should conduct regular security audits to evaluate how data is being handled by third-party developers and internal systems. Continuous monitoring helps identify suspicious activities or misuse of data. Early detection of security issues can prevent large-scale data breaches and protect user privacy.

7.4 Restrict Third-Party Application Permissions

Social media platforms should strictly regulate the permissions granted to external applications. Developers should not be able to access the data of users' friends without their consent. Implementing stricter permission systems ensures that personal information is shared only with trusted and verified applications.

7.5 Strong Data Protection Regulations

Governments should enforce strict privacy regulations such as the General Data Protection Regulation. These regulations require organizations to follow clear rules for collecting, storing, and processing personal data. Strong legal frameworks encourage companies to adopt responsible data management practices.

7.6 Improved User Privacy Controls

Platforms should provide users with better privacy settings and tools to manage their personal information. Users should be able to easily check what data is collected, which applications have access to their information, and how it is used. Giving users more control over their data helps reduce the risk of unauthorized access and misuse.

8 Conclusion

The Facebook–Cambridge Analytica scandal remains one of the most significant data privacy controversies in the digital era. The incident revealed how personal data from millions of users on Facebook was collected and used without proper consent. The

misuse of data through a third-party application highlighted major weaknesses in data governance, access control, and privacy protection within social media platforms.

The case demonstrated that even without traditional hacking, personal data can still be misused when organizations fail to properly regulate third-party access to user information. The data collected through the application developed by Aleksandr Kogan was later used by Cambridge Analytica for political profiling and targeted advertising. This raised serious concerns about user consent, transparency, and the ethical use of personal data in political campaigns and digital platforms.

The scandal also led to global discussions about the importance of data protection and responsible handling of personal information. Governments and regulatory authorities began to introduce stricter data privacy regulations and policies to ensure that organizations manage user data more responsibly. Technology companies were encouraged to strengthen their privacy controls, restrict third-party access to sensitive information, and improve transparency regarding how user data is collected and used.

Overall, the Facebook–Cambridge Analytica case highlights the need for stronger data governance, ethical data practices, and improved security measures. Organizations must prioritize user privacy, implement strict data protection policies, and ensure transparency in data handling processes to prevent similar incidents in the future and maintain public trust in digital platforms.

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